

UMM E FARWA

Civil Engineer

📍 Karachi, Pakistan

☎ (+92) 330 661 1994

✉ ummay.farwa@outlook.com

🌐 LinkedIn

📁 Portfolio

ABOUT ME

B.E. Civil Engineer with a **CGPA 3.55/4.0**, specializing in structural engineering. Actively engaged in research on ultra-high-performance concrete (UHPC) and the application of artificial intelligence, particularly artificial neural networks (ANN), for advanced construction materials.

Strong interest in structural systems, structural mechanics, and innovative data-driven approaches in Civil Engineering. Intends to pursue a Master's degree focused on intelligent and resilient infrastructure systems.

EDUCATION

Bachelor of Engineering (BE - Civil Engineering)

NED University of Engineering & Technology Karachi Pakistan

Oct,2022 – Jun,2026

Field(s) of study: Civil Engineering – Final grade: CGPA 3.55/4.0 – Level in EQF: 6 – Total credits Hours: 137

Key Coursework: Structural Analysis I & II, Reinforced Concrete Design I & II, Numerical Analysis, Soil Mechanics I & II, Transportation I & II and Construction Project Management.

Intermediate – Higher Secondary School Certificate (HSSC)

St. Lawrance's Govt. Girls Degree College Karachi Pakistan

Oct,2020 – May,2022

Field of study: Pre-Engineering – Marks Obtd/Total: 940/1100 – Percentage: 85% – Grade: A-1 – EQF level: 4

Major Subjects: Physics, Chemistry, Mathematics & English

Matriculation – Secondary School Certificate (SSC)

Jinnah Foundation Secondary School Karachi Pakistan

Apr,2018 – Mar,2020

Field of study: Science – Marks Obtd/Total: 733/850 – Percentage: 86% – Grade: A-1 – EQF level: 3

INTERNSHIPS EXPERIENCE

Paragon Constructor Private Limited (EMAAR Project)

Site Engineer Internee – Karachi, Pakistan – 1 month

Apr,2025

- Conducted daily **site inspections** to monitor work quality, ensure adherence to safety regulations, and verify compliance with approved structural drawings and specifications.
- Assisted in foundational activities, including layout, boring, and concreting for **pile foundations**, emphasizing precision and quality control.
- Prepared detailed **Reinforced Concrete (RC) element schedules** and **Bar Bending Schedules (BBS)** for beams, columns, and slabs.

AAA Partnership Private Limited

BIM Modeler Internee – Karachi, Pakistan – 1 month

Jan,2025

- Modelled a 400 sq. yard residential house in **Autodesk Revit**, achieving accurate 3D architectural and structural representations.
- Applied **Building Information Modelling (BIM) standards** for model organization and coordinated between CAD drawings and the Revit model to ensure design integrity.
- Completed training in project management using **Procore**, gaining experience in document control and project communication

TECHNICAL & COMPUTATIONAL SKILLS

- Structural / FEA:** ETABS, structural analysis, moment distribution, finite element method (Skilled)
- BIM & Drafting:** Autodesk Revit, Autodesk AutoCAD, reinforcement detailing (Advanced)
- Programming / AI:** Python, MATLAB, ANN-based modeling, computational logic (Intermediate)
- Microsoft Office:** MS PowerPoint, MS Word, MS Excel, MS Project (Advanced)
- Project Management:** Procore training, work breakdown structures (WBS), construction costing (Beginner)
- Laboratory Skills:** CTM/UTM operation, concrete mix design, aggregate testing, tensile testing (Beginner)

RESEARCH INTEREST

- **Computational & Structural Mechanics:** Constitutive modeling of advanced materials, Finite Element Analysis (FEA), structural analysis and design, and complex non-linear material behavior.
- **Artificial Intelligence:** Application of Artificial Neural Networks (ANN) and Machine Learning (ML) for predictive modeling and data-driven design.
- **Advanced & Sustainable Materials:** High and Ultra-High-Strength Concrete (HSC/UHSC), advanced cementitious materials, sustainable construction materials, and their applications in high-performance structural systems.

PROJECTS

Constitutive Modelling of Advanced Concrete using AI Neural Networks

Final Year Design Project – NED University

2025-26

- **Scope:** Focused on developing a high-fidelity, data-driven constitutive model to predict the complex non-linear behavior of **High and Ultra-High-Strength Concrete (HSC/UHSC)**.
- **Methodology:** Leveraged **Artificial Neural Networks (ANN)** and advanced computational tools to process experimental material data, moving beyond traditional empirical constitutive laws.
- **Impact:** The **ANN-based model** is constructed to estimate the key mechanical parameters of UHPC, supporting experimental analysis and optimization of mix design.

Structural Analysis & Design of Academic Block N1

Academic Project – NED University

2024

- Designed the main structural elements (beams, columns, slabs, footings) for an Academic Block using **manual calculations, Moment Distribution Method, Finite Element Approach**, and validated the results using ETABS.
- Produced detailed structural drawings in **AutoCAD** and 3D models in **Revit** for comprehensive representation and detailing.

Python Based Structural Analysis Tool Development

Academic Project – NED University

2025

- **Developed a Python-based application with a Graphical User Interface (GUI)** for the design and analysis of continuous beams in accordance with **ACI-318** standards.
- Applied computational logic to automate tedious design checks, demonstrating proficiency in both structural code and programming logic.

Experimental and Laboratory Work

Materials Testing Laboratory – NED University

2022-25

- Executed **concrete mix design** and performed essential material quality control tests, including slump test, water absorption, and aggregate impact value.
- Conducted **compression testing** on cubes and beams (CTM/UTM) and performed **tensile strength testing of steel** (UTM) to gather data for material performance evaluation.

HONORS & CERTIFICATIONS

- **Certification :** Ultra High Strength Concrete - From Brittle to Impact Resistant (HEC NRPU). Feb ,2026
- **Prime Minister's Youth Laptop Scheme :** Awarded Laptop based on academic performance. Oct, 2025
- **Certification :** Completed training in project management software **PROCORE**. Dec, 2024
- **Competition :** Participated in the **ACI BAMBOOZLE** concrete beam design competition. Jun, 2024
- **Contest:** Participated in **International Kangaroo Linguistic Contest**. Jun, 2017

CONFERENCES & SEMINARS

- **Seminar Attendance:** Attended a seminar on the impact of **Artificial Intelligence** in aviation, with a focus on computational applications. Feb, 2024
- Attended a seminar on **Geo-Synthetic Case Studies** held at the NED University Auditorium. Aug, 2025
- Developed and Presentations **DIY projector** using a magnifying glass on School level Exhibition. Mar, 2017

MEMBERSHIPS

- Member of the American Society of Civil Engineers (ASCE) 2026
- Member of the American Concrete Institute (ACI) 2026
- Member of the Structural Engineering Institute of ASCE 2026